



UNIVERSITEIT VAN AMSTERDAM

Philosophy of Physics

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Contents

1	Quantum Foundations and Quantum Information	1
1.1	The EPR–Bohr Debate	1
1.2	Bell’s Theorem	6
1.3	Quantum Information Theory	9
1.4	Three Views on the Philosophy of Information	12
2	Spacetime	13
2.1	The Equivalence Principles	13
2.2	Energy and Noether’s Theorem in General Relativity	15
2.3	Time and the Hole Argument	16
2.4	Black Holes and Quantum Information	17
3	Dualities and Emergence	18
3.1	Dualities in Physics	18
	References	19

Quantum Foundations and Quantum Information

SECTION 1.1

| The EPR–Bohr Debate

The EPR Argument

Non-commuting quantities:

- values cannot be determined simultaneously;
- if we measure x , then p is indeterminate (has no physical reality).

Entangled state of two spacelike separated particles.

- If measure A for particle 1, know the value for particle 2, and likewise for B .
- Can choose to measure A or B for particle 1. Either way, the values of particle 2 are determined.
- No interaction: Particle 2 must have had values determined beforehand.
- *QM incomplete*, because A and B do not commute (cannot both have values).

EXAMPLE 1.1. Take a state ψ that is an eigenstate of both $P_1 + P_2$ and $Q_1 - Q_2$:

$$(P_1 + P_2)\psi = 0, \quad (Q_1 - Q_2)\psi = \ell\psi,$$

where ψ is not an eigenstate of the individual P 's and Q 's. Measurement of momentum p_1 for particle 1 predicts $-p_1$ for particle 2, and measurement of position a of particle 1 predicts $a - \ell$ for particle 2. *QM incomplete*: Must have had those values already, since measurement of particle 1 cannot affect particle 2. No need to actually carry out the measurements: The possibility of choosing which to measure allows us to make the prediction.

EPR's desideratum for a complete theory of quantum mechanics is as follows.

ASSUMPTION 1.2 (Comp).

If the *formalism of quantum mechanics* (FQM) is complete, then every element of physical reality has a counterpart in FQM.

In other words, nothing needs to be added to FQM for it to describe physical reality.

ASSUMPTION 1.3 (EPR Criterion of Reality).

If, using FQM, we can predict with probability one the result of measuring a physical quantity at time t , then at time t , there is an element of physical reality corresponding to the physical quantity, with a value equal to the predicted result of the measurement.

So, Comp demands that there is *nothing left out* by FQM. And by nothing left out, we mean (EPR-COR): Every quantity that FQM predicts the value of with certainty is encoded in the formalism.

ASSUMPTION 1.4 (EPR-Locality).

Elements of physical reality pertaining to one system cannot be affected by measurements performed on another system that is space-like separated from the first.

EPR's strategy is to describe a thought experiment that denies the consequent of Comp, so that one can deduce that FQM is incomplete. That is, assuming FQM, EPR-COR, and EPR-Locality, they will exhibit a quantity whose value is predicted with certainty by FQM but is nevertheless not encoded in FQM. In this way, they turn the formalism of quantum mechanics against itself.

Version 1

Consider a pair of particles in the following state:

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle_1 |\downarrow_z\rangle_2 - |\downarrow_z\rangle_1 |\uparrow_z\rangle_2). \quad (1.1)$$

For this state, one can derive within FQM:

$$p(\uparrow_{z;2} | \downarrow_{z;1}) = 1 = p(\downarrow_{z;2} | \uparrow_{z;1}), \quad (1.2)$$

i.e., the z -spins of the two particles are perfectly correlated. And note that these relations hold even if particle 1 and particle 2 are space-like separated. Then consider the following argument.

- (i) If at some time t_0 we measure the z -spin of particle 1, we will be able to infer, for all $t > t_0$, the value of the z -spin of particle 2 (with certainty, from (1.2), say).
- (ii) Thus by (i) and EPR-COR, for all $t > t_0$, there is an element of physical reality corresponding to the z -spin of particle 2, with a value equal to minus the value obtained for particle 1.
- (iii) By (ii), and EPR-Locality at t_0 , the z -spin of particle 2 cannot have been affected by the measurement of the z -spin of particle 1 at t_0 . That is, at t_0 , particle 2 already possessed the value of its z -spin deduced in (ii) for $t > t_0$.
- (iv) Thus (since we assume throughout that there are no other influences on the system) particle 2 possessed the value of its z -spin deduced in (ii) for $t < t_0$ also (but, of course, for a time t after the creation of the two particles). That is, there is an element of physical reality, namely, the z -spin of particle 2 (and its value) for $t < t_0$.
- (v) However, nowhere in the state of the system before t_0 , viz. nowhere in the state (1.1), is it manifest that particle 2 has a determinate value for its z -spin. Put another way, prior to the measurement of the z -spin of particle 1, particle 2 was not in an eigenstate of spin in the z -direction. Thus there is an element of physical reality that is not captured by FQM.

(vi) Therefore, FQM is incomplete, i.e., $\neg\text{Comp}$.

In short,

$$\text{FQM} \wedge \text{EPR-COR} \wedge \text{EPR-Locality} \implies \text{FQM is incomplete, i.e., } \neg\text{Comp}.$$

Equivalently,

$$\text{FQM} \implies \neg\text{EPR-COR} \vee \neg\text{EPR-Locality} \vee \neg\text{Comp}.$$

And so we are effectively left with two choices: The first, (i) below, corresponds to EPR's actual conclusion.

- (i) If you accept the formalism of orthodox quantum mechanics (FQM) and you aren't prepared to give up EPR-COR or EPR-Locality, then you are forced to conclude that FQM is incomplete.
- (ii) If you accept FQM and you assume it is complete, and you aren't prepared to give up EPR-COR, then you must give up EPR-Locality.

Version 2

A second version of the argument is closer in spirit to the original argument of EPR. This version argues that from FQM, one can demonstrate the simultaneous existence of definite values for non-commuting quantities. Since non-commuting quantities in FQM cannot simultaneously have definite values, if you are not willing to give up EPR-COR or EPR-Locality, one can deduce, using Comp, that FQM is incomplete.¹ Again, we assume the system is initially in the state $|\text{EPR}\rangle$ as defined in (1.1).

- (i) Assume FQM is complete.
- (ii) Assume that at some time t_0 we measure the z -spin of particle 1, and we find that it has z -spin down. FQM predicts for $t > t_0$ that $p(\uparrow_{z,2} | \downarrow_{z,1}) = 1$. But the particles may be space-like separated, so from FQM and EPR-Locality, $p(\uparrow_{z,2}) = 1$ for all relevant t , including $t < t_0$, but for a time t after the pair was created.
- (iii) From (ii) and EPR-COR, we may infer that particle 2's having z -spin up is an element of physical reality.
- (iv) But we could instead have made an x -spin measurement on particle 1. Suppose we had, and that (without loss of generality) we had obtained x -spin down. FQM predicts that for $t > t_0$, $p(\uparrow_{x,2} | \downarrow_{x,1}) = 1$. But the particles may be space-like separated, so from FQM and EPR-Locality, $p(\uparrow_{x,2}) = 1$ for all relevant t , again including $t < t_0$, but for a time t after the pair was created.
- (v) From (iv) and EPR-COR, we may infer that particle 2's having x -spin up is an element of physical reality.
- (vi) x -spin and z -spin are incompatible quantities; so, given 1, no particle can be in an eigenstate of both simultaneously.
- (vii) But from (iii) and (v), particle 2 is in an eigenstate of x -spin and z -spin simultaneously. From (vi) and (vii): contradiction.

¹ Note that EPR's chosen non-commuting observables were position and momentum, but Bohm, in his 1951 textbook, chose different spin components on the singlet state of two spin-half particles. This has become the standard example, and the one that we are indeed following here.

If we retain EPR-COR and EPR-Locality, as EPR wished to, then we must give up the assumption that FQM is complete.²

Bohr's Reply

EPR mention measuring a physical quantity *without disturbing the system*. But *disturbance* is ambiguous between:

- (i) mechanical disturbance, e.g. by physical interaction (exchange of a particle);
- (ii) an influence on the very conditions which define the possible types of predictions regarding the future behaviour of the system (a disturbance about what we can say about the system).

Bohr agrees that there is no disturbance of (i) because of relativity, but there can be a disturbance of (ii). Applicability of physical concepts depends on *context of measurement*. Experimental circumstances define what you can call physical reality.

Contextualism: cannot say particle 2 has z-spin until we actually measure the z-spin of particle 1. The measurement creates the conditions of possibility under which z-spin can apply, i.e., the conditions under which it is meaningful to talk about z-spin.

- EXAMPLE 1.5. (i) Normal double-slit experiment for fixed slit: state of slit independent of particle's path: *interference fringes visible*. Position does not apply to particle's path.
- (ii) Modified double-slit experiment: State of slit depends on particle's path. Ambiguity in particle's trajectory is removed: *interference suppressed*. Position applies to particle's path.

Bohr's description *leaves some things unclear*: How does measuring a property of particle 1 create the conditions of possibility for particle 2 to have that property?

Anton Zeilinger: *Our choice of measurement apparatus decides which of these quantities can become reality in the experiment.*

Heisenberg: *agreement with relativity, because no information faster than light.*

- Einstein: *Relativity implies no faster-than-light influences, period.*
- Heisenberg: *It is faster-than-light signaling that is relevant.*

Copenhagen interpretation involves *collapse* of the wave function of particle 2. Still mysterious.

Murray Gell-Mann: *Niels Bohr brainwashed a whole generation of theorists into thinking that the job (an adequate philosophical presentation of quantum mechanics) was done fifty years ago.*

Einstein wants observation-independent reality. Bohr is happy with complementary pictures whose applicability depends on measurement context.

- (i) If we measure position, the concept of *position* is applicable. Because of the correlation, *position* is also applicable to the other particle. The measurement apparatus of the first particle defines which classical concepts apply to the second.

² Note, as discussed by Redhead (1987, p. 78), Version 2 of the argument is correct, but it relies on two measurements that cannot be made in a single experiment. Version 1 avoids this complication (and ensuing controversies about interpreting counterfactuals).

- (ii) Essential difference between classical and quantum physics: Classical physics makes no reference to measurement. In quantum physics, the interaction between object and apparatus involves exchanged quanta. The object and measurement apparatus form an irreducible whole.
- (iii) The exchange of momentum cannot be analysed because the momentum is absorbed by fixed parts of the apparatus and leaves no trace.

Recent progress in understanding Bohr's views on complementarity. Two concepts are *complementary* if:

- (i) different meanings: incompatible;
- (ii) *individually insufficient*, but *jointly sufficient*, for a complete description of the system.

EXAMPLE 1.6 (Two Types of Complementarity). (i) *Wave-particle complementarity*: use of wave/particle *pictures* or *models* that correspond to classical notions of waves/particles.

- Can apply to a *single experiment*: Compton scattering.
- Wave and particle properties are *classically incompatible*.

(ii) *Kinematic-dynamic complementarity*: spatiotemporal (x, t) (kinematic) and momentum-energy (p, E) (dynamic) descriptions.

- Requires *different experiments*: fixed (x) /movable (p) setup.
- *Incompatible observables*: not simultaneously well-defined.
- Position and momentum properties are *classically compatible*.

No parallel between (i) and (ii). Bohr's view has recently been advocated in *Algebraic Quantum Field Theory* (AQFT).

SECTION 1.2

| Bell's Theorem

John Bell (1986, 1984) on quantum mechanics: I would like to make a clean theory. And when I look at quantum mechanics, I see that it's a dirty theory. The formulations of quantum mechanics that you find in the books involve dividing the world into an observer and an observed, and you are not told where that division comes—on which side of my spectacles it comes, for example—or at which end of my optic nerve... What you learn... is that, for practical purposes, it does not much matter where you put this division; that the ambiguity is at a level of precision far beyond human capability of testing. So you have a theory which is fundamentally ambiguous... Conventional formulations of quantum theory, and of quantum field theory in particular, are unprofessionally vague and ambiguous.

Bell tentatively accepts the conclusion that FQM is incomplete. Assume FQM can be completed by adding a new variable, called λ , which is some piece of knowledge that is not in FQM. It determines the values of elements of physical reality in EPR-like experiments.

Any such theory is either non-local or has other undesirable features. This conclusion is very general: It is independent of details about collapse or hidden variables.

ASSUMPTION 1.7 (FQM EmpAd).

The formalism of orthodox quantum mechanics is empirically adequate; i.e., the probabilities it predicts are correct (reflected in observed long-run frequencies).

ASSUMPTION 1.8 (Locality).

Conditional on the values of all physical quantities prior to measurement, the probabilities for outcomes of space-like separated measurements factorise.

ASSUMPTION 1.9 (No Conspiracy).

The choices of measurement settings are statistically independent of the values of all physical quantities prior to measurement.

Remark. Conspiracy: Settings are dependent on the values of at least some physical quantities prior to measurement.

The theorem is often described as a *no-go theorem*, i.e., a theorem that rules out a certain class of theories which one would otherwise hope could account for the predictions of FQM.

Hans Reichenbach's *principle of the common cause*: Any correlation between two events A and B ought to be accounted for by a common cause (with a direct causal connection between A and B excluded). So if $p(A, B) \neq p(A)p(B)$, there should exist some C such that:

$$p(A, B | C) = p(A | C)p(B | C).$$

DEFINITION 1.10. (i) $A, B :=$ variables ranging over the possible measurement values 0 and 1.

(ii) $a, b :=$ measurement settings, i.e., *directions* of spin measurement.

(iii) $p(A | a, \lambda) =$ probability of outcome A given measurement setting a and λ .

(iv) A *local hidden variable* λ is a *local common cause* for A and B :

$$p(A, B | a, b, \lambda) = p(A | a, \lambda)p(B | b, \lambda). \quad (1.3)$$

LEMMA 1.11 (Law of Total Probability).

$$p(A, B | a, b) = \sum_{\lambda} p(A, B | a, b, \lambda) p(\lambda | a, b). \quad (1.4)$$

Substituting (1.3) into (1.4) gives *locality*:

$$p(A, B | a, b) = \sum_{\lambda} p(A | a, \lambda) p(B | b, \lambda) p(\lambda | a, b).$$

No conspiracy

$$p(\lambda | a, b) = p(\lambda)$$

is also called the *no retrocausality* or *free will* assumption (Bell). The value of λ is not affected by our choice of measurement parameters a, b , i.e., what we decide to measure on the two sides of the laboratory. A *conspiracy* occurs if λ depends on that choice: *superdeterminism*.

Gerard 't Hooft (early '80s): I raised the question: Suppose that Alice's and Bob's decisions have to be seen as not coming out of free will, but being determined by everything in the theory. John said, well ... that I have to exclude. If it's possible, then what I said doesn't apply. I said, Alice and Bob are making a decision out of a cause. A cause lies in their past and has to be included in the picture.

Anton Zeilinger: We always implicitly assume the freedom of the experimentalist ... This fundamental assumption is essential to doing science. If this were not true, then, I suggest, it would make no sense at all to ask nature questions in an experiment, since then nature could determine what our questions are, and that could guide our questions such that we arrive at a false picture of nature.

Substituting $p(\lambda | a, b) = p(\lambda)$ into (Locality) gives:

$$p(A, B | a, b) = \sum_{\lambda} p(A | a, \lambda) p(B | b, \lambda) p(\lambda) \neq p(A | a) p(B | b).$$

This result, $p(A, B | a, b) \neq p(A | a) p(B | b)$, means that in stochastic hidden-variable theories, the joint probability for outcomes doesn't factorise, and there are correlations.

Locality $\wedge$ No Conspiracy $\implies$ CHSH $\implies \neg$ FQM EmpAd

Assuming *Locality* and *No Conspiracy*:

$$E(a, b) := p(++ | a, b) + p(-- | a, b) - p(+- | a, b) - p(-+ | a, b),$$

we derive *Clauser-Horne-Shimony-Holt* (CHSH):

$$|E(a, b) \pm E(a, b')| + |E(a', b) \mp E(a', b')| \leq 2,$$

which must be satisfied by correlations in EPR-like experiments. By comparison with FQM, one can show that the state $|\text{EPR}\rangle$ yields:

$$p(++ | a, b) = \frac{1}{2} \sin^2 \frac{\theta}{2}, \quad p(+- | a, b) = \frac{1}{2} \cos^2 \frac{\theta}{2},$$

where θ is the angle between a and b , so that $E(a, b) = -\cos \theta$. Take $\theta(a, b) = 0$, $\theta(a, b') = 60^\circ$, $\theta(a', b) = 60^\circ$, $\theta(a', b') = 120^\circ$:

$$|E(a, b) \pm E(a, b')| + |E(a', b) \mp E(a', b')| = \frac{5}{2} > 2.$$

Thus FQM violates the CHSH (Bell) inequalities and stochastic hidden-variable theories are not empirically adequate.

fQM EmpAd $\implies \neg$ CHSH $\implies \neg$ Locality $\vee$ Conspiracy

If we exclude superdeterminism, i.e., uphold *No-Conspiracy*, the conclusion of Bell's theorem is that whatever theory is consistent with predictions of QM must be *non-local*.

QM is itself *non-local* in Bell's sense. This is because there are correlations that depend on how measurement settings are chosen, at space-like separations. Thus there are faster-than-light influences (recall Bohr's *semantic non-locality*) but no faster-than-light interactions.

Neither pilot-wave theory nor GRW theory falls under the assumptions of Bell's theorem. Pilot-wave theory is not a local theory; it includes faster-than-light interactions and therefore does not fall under the assumptions of Bell's theorem. GRW spontaneous-collapse theories deny the existence of hidden variables: They are fundamentally indeterministic.

Remark. (i) For a long time, the orthodoxy was that Bohr won the debate from EPR.

- Bohr's own solution of EPR seems unsatisfactory: At least it needs further development.
- Recent historical and philosophical scholarship has clarified Bohr's position, which has been advocated in the context of AQFT.
- This prompted Bell to look into hidden variables.

(ii) Under fairly general and reasonable assumptions, Bell derives an inequality that local hidden-variable theories must satisfy. This inequality is violated by quantum mechanics. Since quantum mechanics is empirically adequate, these kinds of hidden-variable theories are ruled out.

SECTION 1.3

| Quantum Information Theory

Black hole information paradox (Hawking's 1976 insight): To lose information is to violate the laws of quantum mechanics. A second level of uncertainty on top of Heisenberg's uncertainty principle. It is reasonable to expect that the resolution will involve *quantum information theory (QIT) principles*.

| Wheeler: *It from Bit* (information as fundamental entity in physics similar to energy).

Extravagant claim based on a confusion:

- (i) information as a technical term, e.g. as used in physics à la Shannon. Measures of quantity of information. Information transmission. Technical notion specified using a mathematical and physical vocabulary.
- (ii) everyday concept of information: knowledge, language, meaning. Depends on reader/hearer, language framework/linguistic community. Wittgenstein: language games. Physics has nothing to say about the everyday concept.

Quantum information and classical Shannon information are *species of a single genus*. Shannon information characterises an *information source* and the requirements for *reproducibility/transmission*.

Logical status: Information is an *abstract noun*, not a thing. Abstract nouns do not denote entities with location in spacetime. Information is to be explained in terms of the verb *inform*.

- (i) To contain information is to be in a certain state. *A* contains information about *B* because correlations allow inferences about *B* from knowledge of *A*. Shannon information is closest to containing information.
- (ii) To possess information is to possess a certain ability (as with knowledge).

Linguistic status:

- *token*: concrete thing, existing in spacetime;
- *type*: abstract, not concrete thing. Different tokens for same type.

A sentence is a grammatically well-formed string of words.

Shannon information_t: sequence of symbols. Sound waves, a pattern of amplitudes over time. Sentences and propositions are irrelevant. The *sequence type* is transmitted through the production of a sequence token of the same type.

| Shannon (1948): *Semantic aspects of communication are irrelevant to the engineering problem.*

Alphabet of letters $x_1, x_2, \dots$ which occur with probabilities $p(x_i)$. Message: sequence of N symbols/states, $x_4 x_6 x_1 x_3 \dots$. 2^{NH} distinct typical such messages (sequences of letters, for large N), where

$$H = - \sum_{i=1}^n p(x_i) \log p(x_i).$$

Can code these messages using NH bits.

Shannon's coding theorem: the optimal compression. Information_t per letter is H bits. Information_t transmits the sequence type.

Shannon Information_t is about transmission of a sequence of symbols. Nothing to do with information, learning, or the aim of the transmission.

Levels of everyday concept: *syntactic* (sentence token, sentence type), and *semantic* meaning (proposition). Shannon theory is about none of these, only sequence tokens and sequence types.

Perhaps the analogy between sequences of mathematical symbols and linguistic sentences is stronger than Timpson suggests? Nevertheless, the main point is that meanings are irrelevant to Shannon information theory.

An extension of Shannon Information_t: quantum information. The quantum analogue of H is von Neumann entropy:

$$S(\rho) := -\text{Tr} \rho \log \rho = - \sum_{i=1}^n \lambda_i \log \lambda_i.$$

Important difference: no-cloning theorem. Q-information cannot be copied. Linearity of quantum mechanics.

As in the classical case, quantum information_t has nothing to do with uncertainty or learning about the system. What is transmitted is a *sequence of quantum states*. Production of token of sequence type. Source may be mixed state ρ , may include entangled parts, and parts not visible at destination.

An item of quantum information_t is a particular *sequence of Hilbert space states*. Just like Shannon information, it is an abstract quantity, not a physical object. It is not information_t, but its tokens, that are physical.

Remark. (i) *Information and information_t are both abstracta: properties of things, not things themselves. The token-type distinction explains the transmission of information.*

(ii) *Everyday information is semantic (about meanings). Information_t is merely about sequences/states, not meanings.*

(iii) *Information and uncertainty conceptually distinct.*

(iv) *Quantum information_t is conceptually the same as Shannon information. Abstract, not physical.*

EXAMPLE 1.12 (Quantum Teleportation). A case in which information is physical is counter-intuitive: Treating information_t as abstract dissolves puzzles.

(i) Alice has an unknown initial spin- $\frac{1}{2}$ state $|\chi\rangle_1$:

$$|\chi\rangle_1 |\text{EPR}\rangle_{23} = \frac{1}{\sqrt{2}} (\alpha \uparrow_1 + \beta \downarrow_1) (\uparrow_2 \downarrow_3 - \downarrow_2 \uparrow_3).$$

By straightforward algebra, rewrite this as

$$|\chi\rangle_1 |\text{EPR}\rangle_{23} = -\frac{1}{2} |\text{EPR}\rangle_{12} |\chi\rangle_3 + (\text{terms of similar form}). \quad (1.5)$$

All terms contain a copy of the initial unknown state $|\chi\rangle_3$.

(ii) Alice measures systems 1 and 2 in the Bell basis and finds one of four terms in (1.5). If her state collapses³ to $|\text{EPR}\rangle_{12}$, only the first term in (1.5) is left: System 3 is in state $|\chi\rangle_3$.

³ State $|\chi\rangle_1$ disappears at Alice's place (no-cloning) after she measures.

Likewise, the other summands in (1.5) all contain $A|\chi\rangle_3$, with A a unitary operator. Alice sends a message to Bob to apply $A^\dagger$ to his state (no transmission of quantum states, only two classical bits).

- (iii) After the application of $A^\dagger$, Bob performs an operation on his state and gets a copy of the initial unknown state $|\chi\rangle_3$.

Remark. (i) No matter is transported: only the quantum state $|x\rangle$, i.e., a piece of quantum information_t.

(ii) No transmission faster than light: Bob waits for Alice's two bits, at speed $< c$.

(iii) No question of how information_t is transferred. Information_t is a property of Alice's system, and then of Bob's system.

(iv) No thing travelling from A to B: no question about transmission, beyond description of physical process in protocol.

Jozsa and Penrose: Information travels backwards in time from A's measurement to time when entanglement created, then forward in time.

(v) No such outlandish solution needed: puzzles dissolved.

SECTION 1.4

| Three Views on the Philosophy of Information

Floridi's Philosophy of Information

Information is well-formed, meaningful, truthful data. Semantic view of information. Data (bits, qubits) are uninterpreted. False information not a kind of information, but misinformation.

Information thus understood casts light on philosophical problems: knowledge, reality and ethics (approaches to, e.g., informational privacy, the intrinsic value of the infosphere, distributed ethics, and the ethics of artificial agents).

Informational Structural Realism

Worrall (1989): *There was an important element of continuity in the shift from Fresnel to Maxwell... There was continuity or accumulation in the shift, but the continuity is one of form or structure, not of content.*

- (i) *Structural realism*: Scientific theories tell us only about the mathematical structure of the world (quantitative relations between objects), not about the nature of objects.
- (ii) *Epistemic structural realism*: Scientific theories tell us about (mathematical) relations between objects. Suspend judgement as to their nature.
- (iii) *Ontic structural realism*: There are no objects/things, structure is all there is.
- (iv) *Informational structural realism*: Ultimate nature of reality is informational. Reality is constituted by structural objects, i.e., coherent clusters of data.

Digital Ontology

Aristotle (*metaphysics*): *The so-called Pythagoreans, who were the first to take up mathematics, not only advanced this subject, but saturated with it, they fancied that the principles of mathematics were the principles of all things.*

Digital ontologists (e.g., Konrad Zuse): The nature of reality is digital and discrete; the universe is a big Turing machine.

John Wheeler: *It from bit symbolizes the idea that every item of the physical world as at bottom ... an immaterial source and explanation; that which we call reality arises in the last analysis from the posing of yes-no questions ... in short that all things physical are information-theoretic in origin and this is a participatory universe.*

Spacetime

SECTION 2.1

| The Equivalence Principles

Einstein Equivalence Principle

In free fall, there is no gravitational field. Gravitation (acceleration in fall) and acceleration (intensity of gravitational field) are the same. *Einstein equivalence principle (EEP)*: inertia = gravitation.

An observer has no way to distinguish between being uniformly accelerated but not subject to gravity and being at rest in a homogeneous gravitational field. As in Galileo's example: an inability to distinguish motion from rest.

Weak Equivalence Principle

- *Newton's second law of motion*: gravitational force exerted by Earth = inertial mass $\times$ acceleration in fall: $F = m_i a$.
- *Newton's gravitational law*: gravitational force exerted by Earth = gravitational mass $\times$ intensity of the gravitational field:

$$F = \frac{GMm_g}{R^2} = m_g \left(\frac{GM}{R^2} \right) = m_g g.$$

- *EEP*: acceleration in fall = intensity of the gravitational field: $a = g$ (or inertia = gravitation in the Newtonian limit).
- *Weak equivalence principle*: inertial mass = gravitational mass.

Confirmed by Eötvös's experiments: $m_i = m_g$. The ratio $\frac{m_i}{m_g}$ is independent of the material.

Strong Equivalence Principle

Einstein (1920): The fact from which we started out, namely that no gravitational field exists in the immediate vicinity of a free-falling observer, this very fact shows that in the vicinity of every world point the results of the theory of special relativity are valid (in the infinitesimal) for a suitably chosen local coordinate system.

The SEP is a consequence of the EEP. GR is locally special relativistic. Locally there is no evidence of gravity. For infinitesimal regions, SR holds for suitable coordinates. The components of the affine connection are zero in those coordinates.

SECTION 2.2

| Energy and Noether's Theorem in General Relativity

SECTION 2.3

| Time and the Hole Argument

SECTION 2.4

| Black Holes and Quantum Information

Dualities and Emergence

SECTION 3.1

| Dualities in Physics

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